

Kevin's Vim IDE

Setup Reference · Pathogen · Catppuccin Mocha · Node/Express · .NET (C#/Blazor) · Python · Next.js

This Vim configuration turns Vim into an IDE targeting parity with VS Code for the stacks used day to day: Node.js/Express, ASP.NET Core (C# and Blazor), Python, and Next.js. Language intelligence (autocomplete, go-to-definition, go-to-implementation, hover docs, diagnostics, and fixers) is provided by ALE talking LSP to pylsp (Python) and typescript-language-server (JS/TS/JSX/TSX), and by omnisharp-vim talking directly to the OmniSharp/Roslyn server for C#. The plugin manager is pathogen.vim: each plugin lives as its own directory under ~/.vim/bundle/, added with a plain `git clone` and no separate lockfile or build step.

Source of truth for VIM_IDE_Setup.pdf. This Markdown file is the editable, up-to-date reference — regenerate the PDF with `pandoc VIM_IDE_Setup.md -o VIM_IDE_Setup.pdf --pdf-engine=weasyprint`.

Plugin Manager — pathogen.vim

execute `pathogen#infect()` at the top of `.vimrc` adds every directory in `~/.vim/bundle/*` to Vim's runtimepath. Installing a plugin is `git clone <url> ~/.vim/bundle/<name>`; updating is `git pull` inside that directory. `install-vim-ide.sh` (included alongside this document) automates both for the full plugin list below.

Installed Plugins

Plugin	Repository	Purpose
ale	dense-analysis/ale	Async Lint Engine — LSP client, diagnostics, fixers (black/isort/prettier), completion for Python/JS/TS

Plugin	Repository	Purpose
omnisharp-vim	OmniSharp/ omnisharp-vim	C# completion, go-to-definition/ implementation, hover, docs via OmniSharp/Roslyn
vim-razor	jlcrochet/vim-razor	Syntax highlighting + indent for .razor / .cshtml (Blazor markup)
supertab	ervandew/supertab	Tab-triggered context-aware completion (routes to ALE/OmniSharp omnifunc)
nerdtree	preservim/nerdtree	File tree explorer sidebar
nerdtree-git-plugin	Xuyuanp/nerdtree- git-plugin	Git status flags (modified/staged/ untracked/...) inline in the file tree
vim-nerdtree-syntax-highlight	tiagofumo/vim- nerdtree-syntax- highlight	Per-filetype color highlighting of names/icons in the file tree
vim-devicons	ryanoasis/vim- devicons	Filetype icons in NERDTree, airline and statusline (needs a Nerd Font)
vim-fugitive	tpope/vim-fugitive	Git integration — :Git status/diff/ blame/commit/push, inline blame
vim-gitgutter	airblade/vim- gitgutter	Git change markers (+/-/~) in the sign column, stage/undo hunks
vim-airline	vim-airline/vim- airline	Statusline/tabline
vim-airline-themes	vim-airline/vim- airline-themes	Airline theme pack

Plugin	Repository	Purpose
catppuccin-vim	catppuccin/vim	Catppuccin Mocha colorscheme + matching airline theme
fzf-vim	junegunn/fzf.vim	Fuzzy file/buffer/tag finder (:FZF)
tagbar	preservim/tagbar	Code outline / symbol sidebar (ctags-based), <F8>
asyncrun-vim	skywind3000/asyncrun.vim	Background shell commands (dotnet/ npm/etc.) into the quickfix window
vim-polyglot	sheerun/vim-polyglot	Broad syntax highlighting for many languages/ filetypes
indentLine	Yggdroot/indentLine	Vertical indent guides
vim-illuminate	RRethy/vim-illuminate	Highlights other occurrences of the word under the cursor
vim-surround	tpope/vim-surround	Add/change/delete surrounding quotes, tags, brackets
vim-commentary	tpope/vim-commentary	Toggle line/block comments (gcc / gc)
vim-visual-multi	mg979/vim-visual-multi	Multiple cursors (VS Code Ctrl+D equivalent)
vim-startify	mhinz/vim-startify	Start screen — recent files/sessions
vim-auto-save	907th/vim-auto-save	Autosaves on change, like VS Code's Auto Save
emmet-vim	mattn/emmet-vim	Emmet abbreviation expansion for HTML/ JSX/CSS

Plugin	Repository	Purpose
vim-tmux-navigator	christoomey/vim-tmux-navigator	Seamless Ctrl+hjkl movement between Vim splits and tmux panes

Theme — Catppuccin Mocha

Colorscheme: `colorscheme catppuccin_mocha` with `termguicolors` on.
 Airline: `g:airline_theme='catppuccin'`, powerline separators and a
 tabline showing open buffers. Icons (NERDTree, airline, tabline)
 require a Nerd Font — this machine already has JetBrainsMono Nerd
 Font installed and available.

Core Keybindings

Leader key is `<Space>`.

Key	Action
<code><leader>w / q / x / Q</code>	Save / quit / save+quit / force-quit-all
<code><leader>e</code>	Toggle NERDTree
<code><leader>f</code>	Reveal current file in NERDTree
<code><leader>ff</code>	Fuzzy-find files (FZF)
<code>Esc Esc</code>	Clear search highlight
<code><F2></code>	Toggle paste mode
<code><F8></code>	Toggle Tagbar (code outline)
<code><leader>rn</code>	Rename symbol (ALERename)
<code><leader>ca</code>	Code action (ALECodeAction)
<code>[e /]e</code>	Jump to previous / next diagnostic
<code>]t / [t</code>	Next / previous tab

Code Navigation & LSP

Key	Action
F12	

Key	Action
	Go to definition (ALE → pylsp / tsserver; OmniSharp for C#)
Shift-F12	Go to definition in a vertical split
F11	Go to implementation
gd / gD	Go to definition / in vertical split (same engine as F12)
gr	Find references
K	Hover documentation (floating preview)
Tab (insert mode)	Context-aware completion (SuperTab → omnifunc)
. (insert mode)	Auto-triggers omni-completion after member access

Git Integration

vim-fugitive provides full Git commands from inside Vim; vim-gitgutter shows per-line change markers in the sign column and can stage/undo individual hunks; nerdtree-git-plugin mirrors file status as flags directly in the file tree.

Key	Command
<leader>gs	:Git status
<leader>gd	:Git diff
<leader>gb	:Git blame
<leader>gc	:Git commit
<leader>gp	:Git push

Per-Language Setup

Node.js / Express

Filetypes: javascript, javascriptreact, typescript, typescriptreact.
Completion/hover/go-to-def via ALE → typescript-language-server; lint via eslint; format-on-save via prettier. :NodeExpressSetup (<leader>ns) writes a jsconfig.json and installs @types/node + @types/express so plain Express JS gets real type-aware completion.

Key / Command	Action
F9	node % — run current file
F10	npm start
F7	npm test
<leader>ni	npm install
<leader>nr	:NpmRun <script>
<leader>ns	:NodeExpressSetup — scaffold jsconfig.json + @types packages

Next.js

Covered by the same typescriptreact/javascriptreact pipeline above — Next projects ship their own tsconfig.json, which typescript-language-server picks up automatically for full type-aware completion across pages, API routes and components.

.NET — C# and Blazor

Filetype cs: completion, go-to-definition/implementation, hover and documentation come from omnisharp-vim talking to the OmniSharp/Roslyn server (installed via :OmniSharpInstall). Filetype razor (.razor / .cshtml, via vim-razor): syntax highlighting and indent only — Vim has no Razor markup LSP (that exists only in Neovim via roslyn.nvim + rzls.nvim). Code-behind (.razor.cs) is plain C# and gets full OmniSharp support.

Key	Action
F9	dotnet run
F10	dotnet build
F7	dotnet test

Key	Action
gd	OmniSharpGotoDefinition
K	OmniSharpDocumentation

Python

Completion/hover/go-to-def via ALE → pylsp, with jedi (completion, definition, hover, references, symbols), rope (completion, rename) and mypy enabled. Format-on-save runs isort then black. pylsp auto-detects a project virtualenv by walking up from the current file (.venv, venv, env, ...).

Key	Action
F9	python % — run current file

NERDTree — Enhanced File Tree

Hidden files shown, minimal UI, directory arrows. On top of the base plugin: nerdtree-git-plugin adds per-entry git status glyphs (modified/staged/untracked/renamed/unmerged/deleted/ignored), vim-nerdtree-syntax-highlight colors names by filetype, and vim-devicons adds filetype icons — together this matches VS Code's Explorer pane (icons + inline git decorations).

These are NERDTree's own default keys (unmodified in .vimrc), active while the cursor is inside the NERDTree window:

Key	Action
<CR> / o	Open file, or expand/collapse directory
t	Open file in a new tab
T	Open file in a new tab, silently (stays on current tab)
i	Open file in a horizontal split
s	Open file in a vertical split
go	Preview file (open, keep cursor in NERDTree)
O	Recursively open the directory under the cursor
x	Close the parent directory of the current node
X	Close all child directories of the current node
u	Move root up one directory

Key	Action
U	Move root up one directory, keep the old root open
C	Change the tree root to the selected directory
cd	Change Vim's working directory to the selected directory
r	Refresh the selected directory
R	Refresh the entire tree from the current root
m	Open the filesystem menu (add/move/copy/delete)
I	Toggle hidden files
f	Toggle the file filter
F	Toggle whether files are shown (dirs only)
B	Toggle the bookmarks table
?	Toggle the quick-help panel
q	Close the NERDTree window

Reproducing This Setup

Two files travel together: `install-vim-ide.sh` and `vimrc`. Run the script from the same directory as both:

```
bash install-vim-ide.sh
```

It installs system packages (Node, .NET SDK, Python, ctags, ripgrep, fd), global npm tools (typescript-language-server, eslint, prettier), pylsp with its plugins, bootstraps pathogen, clones every plugin listed above into `~/.vim/bundle`, deploys the `vimrc`, and triggers the OmniSharp server install. It is safe to re-run: existing plugin clones are updated with `git pull` instead of re-cloned.

A Nerd Font (e.g. JetBrainsMono Nerd Font) must be installed separately for icons/glyphs to render — the script warns if none is detected but does not install one automatically, since font installation is font-manager/distro specific.